

Differential Diagnosis (Fig. 79-7)

This patient has blepharitis, conjunctivitis, and keratitis.

Elastosis may be present. Similar features are found in the papulopustular subtype of rosacea, but the inflammatory infiltrate also surrounds hair follicles and sebaceous glands. Phymatous rosacea is characterized by prominent elastosis, fibrosis, dermal inflammation, sebaceous hyperplasia, and hypertrophy of sebaceous follicles. Epithelialized tunnels undermine the hyperplastic tissue and are filled with inflammatory debris. *Demodex folliculorum* mites may be found in all types of rosacea within the follicular infundibula and sebaceous ducts.

Systemic Diseases

Systemic conditions that must be differentiated from rosacea include: - Polycythemia vera - Connective tissue disorders (e.g., lupus erythematosus, dermatomyositis) - Carcinoid syndrome - Mastocytosis - Neurologic causes of flushing

Neurologic causes of flushing include: - Brain tumors - Spinal cord lesions - Orthostatic hypotension - Migraine headaches - Parkinson disease

Unilateral auriculotemporal flushing may occur following parotid gland injury or surgery.

Medication-Induced Flushing

Medications associated with flushing include: - All vasodilators - Calcium channel blockers - Nicotinic acid (niacin) - Morphine - Amyl and butyl nitrite - Cholinergic drugs - Bromocriptine - Thyroid-releasing hormone - Tamoxifen - Cyproterone acetate - Systemic steroids - Cyclosporine

The flush caused by nicotinic acid may be blocked with aspirin or indomethacin. Disulfiram, chlorpropamide, metronidazole, phentolamine, and

cephalosporins can induce flushing when combined with alcohol. Amiodarone has been reported to induce rosacea-like eruptions and multiple chalazia.

Other Causes of Flushing

Flushing may also result from: - Food additives such as sulphites, sodium nitrite, nitrates, and monosodium glutamate - Dumping syndrome following gastric surgery, which presents with flushing, sweating, tachycardia, and abdominal pain

Cutaneous Conditions That Mimic Rosacea

Skin disorders that may resemble rosacea include: - Topical steroid-induced acneiform eruption (formerly called steroid-induced rosacea) - Acne vulgaris - Perioral dermatitis - Inflammatory keratosis pilaris - Chronic photodamage

Acne vulgaris and rosacea may coexist, although rosacea typically begins and peaks in later decades, after acne incidence declines. The key distinguishing feature is the presence of open and closed comedones, which are seen in acne vulgaris but not in rosacea.

Rosacea Fulminans

Rosacea fulminans (also known as pyoderma faciale or rosacea conglobata) occurs predominantly in women in their 20s. It presents with sudden onset of confluent papules, pustules, nodules, and draining sinuses on the chin, cheeks, and forehead, against a background of diffuse facial erythema. Its classification remains controversial and it was not recognized as a subtype or variant of rosacea by the NRS Expert Committee.

Perioral Dermatitis

Perioral dermatitis differs from rosacea in distribution, clinical features, and patient demographics. It presents with perioral (and sometimes periorbital) microvesicles, micropustules, scaling, and peeling. It primarily affects younger women and can occur in children. Central facial erythema and inflammatory

papules are not typical. Treatment includes topical and oral antimicrobials. The condition is often exacerbated by topical steroid use.

Steroid-Induced Acneiform Eruption

This condition can mimic papulopustular rosacea. With prolonged use of topical corticosteroids on the face, monomorphic inflammatory papules develop. Management involves discontinuation of the topical steroid and initiation of therapy with oral tetracyclines, topical antimicrobials, topical calcineurin inhibitors, or a combination of these.